
CivicSight

Technical Deep-Dive

Data Pipeline, Feature Engineering & Model Training

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Abstract

Traffic fatalities and serious injuries represent one of the most persistent and preventable public health challenges in American cities. In Austin alone, thousands of vehicle crashes are recorded each year — many of them on the same road segments, year after year, in ways that are strongly predicted by the physical design of those roads. Speed limits, lane configurations, intersection geometry, and the presence or absence of pedestrian infrastructure all measurably shape crash risk. The science is not in dispute. What has remained stubbornly difficult is translating that science into fast, systematic, data-supported action at the city level.

Most municipal transportation departments face the same structural problem: crash records are public, road geometry is public, and the research literature linking the two is extensive — but integrating those datasets into a usable decision-support tool requires engineering investment that planning staff rarely have. Crash data arrives as raw GPS coordinates. Road geometry lives in a separate system. The work of joining them, extracting features, and producing ranked risk inventories has to be rebuilt from scratch for every city that wants it. As a result, infrastructure investment decisions are often driven by complaint volume, political visibility, or incomplete intuition rather than by systematic risk analysis.

CivicSight is a data-driven road safety intelligence platform built for Austin, TX that addresses this gap directly. The system automatically ingests crash records from the City of Austin Open Data Portal, road geometry and attributes from OpenStreetMap, and intersection confirmation from Mapbox — then trains an interpretable statistical model to learn which road characteristics are most strongly associated with elevated crash frequency. Every road segment in Austin receives a continuous safety score. Each score is accompanied by an exact, feature-level breakdown of what is driving it, and a counterfactual simulator lets planners ask "what

would happen to this score if a crosswalk were added here?" — before committing to a formal engineering study.

This document provides a complete technical description of the backend pipeline: how data is sourced and spatially joined, how features are engineered from raw API responses, what the training dataset looks like row by row, how the ridge regression model is fit and why that choice was made, and how safety scores decompose into per-feature contributions that planners can actually interrogate. It also describes the current test suite, discusses what is not yet tested, and outlines how structured A/B validation against Austin's real intervention history would ground the simulator's outputs in observed real-world effects.

Disclaimer. *CivicSight is an academic research prototype. Safety scores are statistical estimates derived from historical public data and do not constitute engineering assessments or official policy recommendations.*

1. Problem Statement

Every year, thousands of vehicle crashes are recorded on Austin's roads. Many result in serious injuries or fatalities — outcomes that decades of traffic safety research have shown to be strongly shaped by the physical design of the road itself. A wide arterial with a 45 mph speed limit and no pedestrian crossings is not dangerous by accident. It is dangerous by design, and that danger is measurable, predictable, and in many cases preventable. Posted speed limits, lane counts, intersection geometry, traffic volume, and the presence or absence of pedestrian infrastructure all independently contribute to crash risk in ways that are well-documented in the engineering literature. The science is not in dispute. What has remained stubbornly difficult is translating that science into fast, systematic, city-scale action.

Austin contains thousands of distinct road segments. A planner trying to decide where to direct the next round of infrastructure investment — a new crosswalk, a speed reduction, a lane reconfiguration — faces a genuine information problem. Crash records exist in the public domain but arrive as raw lists of GPS coordinates and incident codes, not as ranked risk inventories. Road geometry data is similarly available but lives in separate systems with incompatible schemas. Integrating those datasets, extracting meaningful features, and producing spatial risk rankings requires sustained engineering investment that most municipal planning departments cannot sustain alongside their core operational responsibilities. The result is that infrastructure decisions are frequently driven by complaint volume, political visibility, or engineering intuition on individual corridors rather than by any systematic analysis of where risk is actually concentrated across the full road network.

The temporal dimension compounds the problem. Infrastructure budgets are allocated annually, but the analysis needed to justify specific investments is rarely completed before allocation decisions are made. A department that wants to propose five new marked crosswalks for next fiscal year needs to identify which five locations are most likely to produce safety gains — but the data infrastructure to support that ranking typically does not exist in a form that can be queried quickly. By the time a formal traffic study is commissioned and completed, the budget cycle has already closed.

The road segments most in need of safety investment are frequently in neighborhoods that have historically received less of it. A tool that surfaces risk objectively — based on crash data rather than which communities have the loudest advocacy presence — has direct implications for equitable resource allocation.

There is also the question of counterfactual reasoning. The most useful planning question is not "where are crashes happening?" — that can be answered by mapping raw crash records. The more valuable question is "where would a specific infrastructure intervention produce the largest safety improvement?" Answering that requires a model that can score the same location under different assumed conditions and estimate the difference. Today, getting that kind of before/after estimate requires commissioning a formal traffic study — a process that takes months, costs significant resources, and is poorly suited to early-stage priority-setting where

planners need fast, rough comparisons across many candidate locations before committing to any of them.

CivicSight is designed to address each of these gaps. It is not a replacement for qualified traffic engineering judgment or site-specific field investigation — no statistical model trained on observational data should be. It is a first-pass decision support layer: a fast, transparent, data-grounded tool that helps planners identify where risk is concentrated across the full Austin road network, understand which specific road characteristics are driving that risk at each location, and get a rough directional estimate of how targeted infrastructure changes might shift predicted outcomes. The goal is to compress the time between "we have a safety budget" and "here are the most defensible places to spend it" from months to minutes.

1.1 System Overview

CivicSight treats every road segment in Austin as a unit of observation. For each segment, measurable characteristics are extracted from public data sources and a statistical model is trained to identify the relationship between those characteristics and observed crash frequency. Once trained, the model scores any segment in the network and responds to counterfactual queries by varying a single feature while holding all others constant — answering questions like "what would the predicted crash rate be if this segment had a marked crosswalk?"

The full pipeline, from raw API call to scored GeoJSON response, runs in four stages:

- Data ingestion — fetch crash records (Socrata/Austin Open Data), road geometry (OpenStreetMap Overpass), and crossing nodes (OSM) in parallel
- Feature engineering — construct one structured row per road segment and confirmed intersection, with numeric and categorical features
- Model training — fit a ridge regression on log-transformed crash counts using those rows
- Scoring & serving — score every segment, decompose each score into per-feature contributions, and serve results via a REST API

The entire pipeline runs at server startup and results are cached for up to six hours, so subsequent requests are served from memory without re-fetching external APIs.

2. Data Sources

2.1 Crash Records — Austin Open Data Portal

Historical vehicle crash records are fetched from the City of Austin Open Data Portal via the Socrata JSON API (dataset ID: `y2wy-tgr5`).¹ Each record represents one reported crash event and includes a GPS coordinate and a unique incident ID. The system fetches up to 120,000 rows by default. No date filtering is applied — the full available history is used. Each crash is stored as

¹City of Austin Open Data Portal. Socrata dataset `y2wy-tgr5`. <https://data.austintexas.gov>

a lightweight point object with three fields: a unique ID, a latitude, and a longitude. No other attributes (severity, weather, time of day) are currently used — the model treats every crash as a single count event.

Field	Type	Example	Notes
id	string	2024-0091234	Unique crash incident ID from Socrata
lat	number	30.2841	GPS latitude of crash location
lng	number	-97.7431	GPS longitude of crash location

Table 1. Crash point schema after Socrata fetch. Only these three fields are retained.

2.2 Road Geometry — OpenStreetMap Overpass API

Road centerlines and attributes are fetched from OpenStreetMap via the Overpass API.² The query retrieves all features tagged `way["highway"]` within Austin's bounding box, excluding pedestrian-only classifications. Each OSM way carries a numeric ID, a highway classification string (e.g. "primary", "residential"), a list of coordinate pairs forming the road centerline, and a tag dictionary containing optional fields like `maxspeed` and `lanes`.

After fetching, ways are ranked by a priority score that favors major road types and higher traffic proxy values, then capped at 6,500 ways for topology processing and 2,500 ways for final rendering. This cap keeps the spatial join and model training tractable while prioritizing the roads most relevant to safety planning.

Field	Type	Example	Notes
id	number	12345678	OSM way ID
highway	string	primary	Road classification tag
coords	[lng,lat][]	[[-97.74,30.28],...]	Polyline coordinate list
tags.maxspeed	string	45 mph	Optional; parsed to integer
tags.lanes	string	2	Optional; parsed to integer
tags.crossing	string	marked	Optional crosswalk signal

Table 2. OSM way schema from Overpass API. Tag fields are optional and may be absent.

2.3 Crossing Nodes & Intersection Confirmation

A second Overpass query fetches all OSM nodes tagged `highway=crossing` within the same bounding box. These pedestrian crossing points supplement way-level crossing tags when determining whether a segment has a nearby marked crosswalk.

²OpenStreetMap contributors. Overpass API. <https://overpass-api.de>

Intersection candidates are derived from OSM topology — vertices shared by two or more road ways — then confirmed against the Mapbox Streets tileset via the Tilequery REST API.³ A candidate is confirmed only if Mapbox returns at least two distinct road features within a small search radius, filtering out grade-separated crossings and OSM tagging artifacts.

3. Dataset Construction

3.1 Unit of Observation

The training dataset contains one row per road segment or confirmed intersection. Road segments are OSM way LineStrings — each representing a contiguous stretch of road between junctions. Intersections are Point features at confirmed junction coordinates. Both types share the same feature schema and are trained on together; the `is_intersection` column distinguishes them.

Row type	Geometry	Typical count	Crash attribution buffer
Road segment	LineString	~2,500	22 m from centerline
Intersection	Point	~250	42 m radius
Total	—	~2,750	—

Table 3. Dataset composition. All rows are used for training — no train/test split currently.

3.2 Crash Attribution (Spatial Join)

The `accident_count` for each row is computed by counting how many crash GPS points fall within a proximity buffer of that segment or intersection. For road segments, a 22-meter buffer is applied — roughly the width of a two-lane road with shoulders. For intersection points, a 42-meter radius captures crashes occurring in the immediate approach zones on all arms of the junction. Crashes are deduplicated by ID to prevent the same event from being counted twice for overlapping segments.

To make this spatial join tractable at scale, crash points are pre-indexed into a grid of approximately 1.1 km cells. When checking whether a crash falls near a segment, only the grid cells overlapping that segment's bounding box are searched rather than scanning all 120,000 crash records.

3.3 Feature Engineering

Once crash counts are attributed, each row's remaining features are derived from OSM tags and lookup tables. No external ML preprocessing library is used — all feature logic is hand-coded in the build pipeline.

traffic_volume

³Mapbox Tilequery API. <https://docs.mapbox.com/api/maps/tilequery>

There is no real-time or historical traffic count integrated yet. Traffic volume is approximated by a deterministic lookup table keyed on OSM highway class, calibrated to approximate typical Austin AADT ranges.

OSM highway class	Traffic volume proxy (vehicles/day)
motorway	55,000
trunk	40,000
primary	28,000
secondary	16,000
tertiary	12,000
unclassified	9,000
residential	6,000
service	4,500
living_street	3,500
(default / other)	7,500

Table 4. Traffic volume proxy values by OSM highway class. These are deterministic lookups, not measured counts.

speed_limit

The OSM `maxspeed` tag is parsed when present and clamped to [10, 85] mph. When absent, class-based fallbacks apply: motorway → 70, trunk → 55, primary → 45, secondary → 40, tertiary → 35, residential/living_street → 30, all others → 35.

num_lanes

The OSM `lanes` tag is parsed when present and clamped to [1, 8]. Class-based fallbacks: motorway/trunk → 3, primary/secondary → 2, all others → 1.

crosswalk_present

Set to 1 if either of two signals is detected: (1) the OSM way carries a `crossing` tag with a positive value (e.g. `crossing=marked`, `crossing=traffic_signals`, `crossing=zebra`), or (2) an OSM `highway=crossing` node exists within 22 meters of the segment centerline.

urban_density

Computed as $\text{accident_count} / \text{max_accident_count_across_all_segments}$, clamped to [0, 1]. For intersections, the denominator is floored at 30 to prevent outlier intersections from compressing the road segment range. This makes `urban_density` a relative measure of local crash intensity rather than an absolute count.

3.4 Example Training Rows

segment_id	acc.	traffic	speed	lanes	road_type	crosswalk	is_ix	density
osm-way/12345	7	28,000	45	2	primary	0	0	0.41
osm-way/67890	0	6,000	30	1	residential	0	0	0.00
osm-way/11111	2	16,000	40	2	secondary	1	0	0.12
mapbox-ix/-97.74,...	14	28,000	45	2	primary	1	1	0.82
osm-way/22222	0	4,500	35	1	service	0	0	0.00

Table 5. Representative training rows. 'acc.' = accident_count (Y). All other columns form the feature matrix X.

4. Training Target and Feature Matrix

4.1 Y — What the Model Predicts

The prediction target is accident_count: the number of distinct crashes spatially attributed to each segment via the buffer logic in Section 3.2. Raw crash counts are right-skewed — most segments have zero crashes, while a handful of high-risk intersections have dozens. Training directly on raw counts would cause the model to overfit those outliers.

To address this, the model trains on $\log(1 + \text{accident_count})$ rather than the raw count. After predicting, the transformation is reversed via $\exp(\text{prediction}) - 1$ to recover an estimated crash count.

accident_count (raw)	$\log(1 + \text{count})$ — trained target
0	0.000
1	0.693
3	1.386
7	2.079
14	2.708
30	3.434
80	4.394

Table 6. Raw crash count vs. log-transformed training target. The log compression prevents outlier intersections from dominating training.

4.2 X — The Feature Matrix

Each row in X is a vector of length $p = 1 + 4 + 2 + k$, where 1 is a constant bias term, 4 are standardized continuous features, 2 are binary indicators, and $k \leq 12$ are one-hot road-type columns. In a typical Austin build with 12 road type columns, $p = 19$.

Column index	Feature	Type	Transformation
0	intercept	constant	Always 1
1	traffic_volume	continuous	$\log(1+v)$, then z-score
2	speed_limit	continuous	z-score
3	num_lanes	continuous	z-score
4	urban_density	continuous	z-score
5	crosswalk_present	binary	0 or 1, no scaling
6	is_intersection	binary	0 or 1, no scaling
7–18	road_type_*	one-hot	0 or 1 per class, no scaling

Table 7. Feature matrix column layout. $p = 19$ in a typical build.

5. Model Training

5.1 What Training Actually Does

The model finds one coefficient (multiplier) per feature. The prediction for any segment is a weighted sum:

$$\begin{aligned} \text{predicted log-crashes} = & w_0 + w_1 \cdot z(\text{traffic}) + w_2 \cdot z(\text{speed}) + w_3 \cdot z(\text{lanes}) \\ & + w_4 \cdot z(\text{density}) + w_5 \cdot \text{crosswalk} + w_6 \cdot \text{is_intersection} + \sum \text{road_type} \\ & \text{weights} \end{aligned}$$

Training finds the specific set of w values that minimizes total squared prediction error across all ~2,750 training rows. This is solved via a closed-form matrix equation — no iterative search required.

5.2 Ridge Regularization

To prevent any coefficient from growing arbitrarily large — which happens when features are correlated — a penalty term is added that discourages large weights.⁴ The regularization strength starts at $\lambda = 5$. If the solver fails due to numerical ill-conditioning, it retries with $\lambda \in \{50, 500, 5,000, 50,000, 500,000\}$. If all attempts fail, the system falls back to predicting the dataset mean for every segment and flags this condition in the API response. The intercept term w_0 is never penalized — it is free to center predictions correctly.

5.3 Crosswalk Coefficient Constraint

⁴Hoerl & Kennard. Ridge Regression: Biased Estimation for Nonorthogonal Problems. *Technometrics*, 12(1), 1970.

After training, the crosswalk coefficient is checked. Because crosswalks tend to be installed at already-dangerous locations, the training data shows crosswalks correlated with higher crash counts — the wrong direction for a what-if simulator. If the fitted coefficient is positive, it is clamped to -0.18 , corresponding to at least a 16% reduction in predicted log-crashes when a crosswalk is present.

This is a deliberate post-training override. It ensures the simulator gives directionally sensible answers, but means the exact magnitude of the crosswalk effect should be read as a rough indicator rather than a precise causal estimate.

5.4 From Prediction to Safety Score

The model outputs a log-crash prediction. To convert to a crash count: $\text{predicted_crashes} = \max(0, \exp(\log_prediction) - 1)$. To produce the 0–1 safety score shown in the map:

$$\text{safety_score} = \exp(-0.08 \times \text{predicted_crashes})$$

A score of 1.0 = zero predicted crashes. The decay constant 0.08 was calibrated so that the score range is meaningfully spread across Austin's observed crash density distribution.

Predicted crashes	Safety score
0	1.00
2	0.85
5	0.67
10	0.45
20	0.20
30	0.09

Table 8. Predicted crash count → safety score conversion.

6. Score Explainability

Because the model is a weighted sum, every prediction decomposes into exact per-feature contributions. For any segment, the log-space risk is the sum of (coefficient × feature value) for each column. These additive terms are computed and returned in the API response alongside every scored segment, allowing planners to see not just a score but a breakdown of what is driving it.

Feature	Normalized value	Log-risk contribution	Interpretation
Intercept	—	+0.30	Baseline for an average segment
Traffic volume	+1.8 σ	+0.28	Well above average traffic
Speed limit	+1.2 σ	+0.18	Above-average posted speed
Num lanes	+0.4 σ	+0.04	Near average lane count
Urban density	+0.9 σ	+0.12	Moderately crash-dense area
Crosswalk absent	0	+0.09	No crosswalk detected
Is intersection	1	+0.03	Intersection flag active
Road type (primary)	1	+0.18	Primary arterial class
Total log-risk	—	+1.22	→ safety score $\approx$ 0.31

Table 9. Illustrative per-feature contribution breakdown for a high-risk primary arterial intersection.

7. Limitations & Caveats

No train/test split. All rows are used for training. There is no held-out evaluation of generalization performance. Scores should be interpreted as relative risk rankings, not calibrated absolute crash count predictions.

Traffic volume is a proxy. The traffic_volume feature is derived from a lookup table keyed on road class, not from measured TxDOT AADT data. Roads with atypical volumes for their class will be systematically mis-scored.

Crosswalk endogeneity. The crosswalk coefficient is manually clamped post-training to enforce a protective direction. The simulated crosswalk effect is a directional estimate, not a causal treatment effect.

OSM tag completeness. Speed limits, lane counts, and crosswalk tags are missing for many Austin segments. Class-based fallbacks are applied but introduce noise for atypical roads.

Static crash history. The model treats the entire crash record as a single aggregate. Seasonal patterns, long-term trends, and effects of prior interventions are not modeled.

Spatial buffer heuristics. The 22 m and 42 m attribution buffers are heuristic. At complex interchanges, crashes may be attributed to the wrong segment.

8. Testing

8.1 Unit Tests

CivicSight's current test suite targets the model layer directly, focusing on behavioral correctness rather than prediction accuracy. Tests are written against the core modeling functions and run

on synthetically generated road segments so they can execute without any external API calls or live data.

Model stability

The first test verifies that the ridge regression solver completes successfully on a representative set of synthetic segments spanning a realistic range of feature values — varying traffic volumes, speed limits, lane counts, road types, and crash frequencies. It checks that the resulting model object is populated and that the solver did not fall back to the degenerate intercept-only mode, which would indicate numerical failure. This test catches regressions in the training pipeline caused by changes to feature construction or regularization logic.

Crosswalk direction — end-to-end

The second test trains a fresh model on synthetic data and then scores the same probe segment twice: once with a crosswalk absent and once with one present, holding every other feature constant. It asserts that the version with a crosswalk never produces a higher predicted crash count than the version without. This validates that the post-training coefficient constraint is functioning correctly end-to-end — from training through prediction — and that the what-if simulator will always report crosswalk additions as neutral or beneficial, never harmful.

Crosswalk weight sign

The third test checks the same property from a different angle: rather than comparing predictions, it inspects the crosswalk feature's additive contribution in log-space directly and asserts it is non-positive. Where the previous test verifies the observable output behavior, this test verifies the internal model state. Together the two tests catch both the case where the coefficient is wrong and the case where the contribution calculation itself is incorrect.

8.2 What Is Not Tested

The current suite does not evaluate whether the model's predictions are accurate. There is no comparison against held-out crash counts, no cross-validation, and no benchmark against a baseline predictor. The tests establish that the model is internally consistent and directionally correct for the crosswalk feature — they do not establish that it generalizes well to unseen road segments.

8.3 A/B Testing — Future Direction

If this project continued into a production or research deployment, a natural next step would be structured A/B testing on the crosswalk counterfactual simulator. The goal would be to move beyond the current directional check — which only confirms that adding a crosswalk is predicted to help — toward a quantitative validation of whether the predicted effect size is realistic.

The setup would work as follows. Austin has a documented record of crosswalk installations, including locations and approximate dates. For each installation, crash counts before and after the installation date could be extracted from the Socrata crash dataset. That real before/after delta could then be compared against what the model's what-if simulator would have predicted for the same location prior to installation. If the model's predicted improvement is consistently much larger or smaller than the observed real-world change, the decay constant and crosswalk coefficient would need recalibration.

This would not be a randomized controlled experiment — crosswalk locations are chosen by planners, not randomly assigned — so the comparison would still be observational. But it would ground the simulator's outputs in something more concrete than the current coefficient constraint, and would give planners a calibrated sense of how much to discount the predicted score improvements when evaluating potential interventions.

The current unit tests confirm that the model behaves correctly by construction. A/B validation against real intervention histories would confirm that it behaves correctly in practice.

9. Future Work

- Integration of TxDOT AADT traffic volume measurements to replace the OSM-class proxy
- Formal train/test partitioning and cross-validated evaluation with RMSE and MAE reporting
- Extension of the what-if simulator to speed limit changes, lane reductions, and signal timing
- Incorporation of Capital Metro transit stop locations and pedestrian count data
- Structured validation sessions with Austin Department of Transportation staff
- Exploration of Poisson regression and negative binomial GLMs as statistically better-suited alternatives
- Temporal modeling to capture seasonal crash patterns and retrospective intervention effects
- Investigation of instrumental variable or difference-in-differences approaches for causal crosswalk estimates

References

See footnotes throughout the document for inline source citations. Primary references:

- [1] City of Austin Open Data Portal. Vehicle Crash Records. Socrata dataset y2wy-tgr5. <https://data.austintexas.gov>
- [2] OpenStreetMap contributors. Road network data via Overpass API. <https://overpass-api.de>
- [3] Mapbox. Tilequery API. <https://docs.mapbox.com/api/maps/tilequery>
- [4] Hoerl, A.E. and Kennard, R.W. Ridge Regression: Biased Estimation for Nonorthogonal Problems. *Technometrics*, 12(1), 1970.
- [5] Lord, D. and Mannering, F. The statistical analysis of crash-frequency data. *Transportation Research Part A*, 44(5), 2010.